Solve quadratic equations using the quadratic formula

1 The standard form of a quadratic equation is $ax^2 + bx + c = 0$.
Identify the values of a, b, and c in the quadratic equation $-8x^2 + 3x = 0$.

2 Using the standard form of a quadratic equation with $a > 0$, identify the values of a, b, and c in the following quadratic equations.

a $x^2 + 7x + 10 = 0$

b $x^2 - 3x - 4 = 0$

c $2x^2 + 9x = 0$

d $8x^2 - 3x = 0$

e $4x^2 + 3x = 5$

f $-2x^2 + 9x + 5 = 0$

g $3x^2 - 8x + 2 = 9x - 7$

3 Find the solutions to the quadratic equation of the form $ax^2 + bx + c = 0$, with $a > 0$.
Express your answer in terms of a, b, and c.

4 To solve $x^2 - x = 2$, Emma substitutes $a = 1$ into the quadratic formula.
Find the value of c that she must substitute into the quadratic formula.

5 Consider the quadratic equation $9y^2 = 8y$. Suppose $b = 8$ in the quadratic formula.

a State the value of a.

b State the value of c.

6 Solve the following equations using the quadratic formula, leaving your answers in exact form.

a $x^2 + 11x + 28 = 0$

b $x^2 - 5x + 6 = 0$

c $4x^2 + 6x + 2 = 0$

d $4x^2 - 7x - 15 = 0$

e $x^2 + 5x + \frac{9}{4} = 0$

f $-10 + 23x + 5x^2 = 0$

g $x(x - 4) = 5$

h $-6 - 13x + 5x^2 = 0$

i $x^2 - 5x - 2 = 0$

j $4x^2 - x - 10 = 0$

k $-2x^2 - 15x - 4 = 0$

7 Solve the following equations using the quadratic formula, rounding your answers to two decimal places.

a $x^2 + 7x - 3 = 0$

b $5x^2 + 9x + 2 = 0$

c $1.8x^2 + 5.2x - 2.3 = 0$

d $3x(x + 4) = -3x + 4$

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8 Using the quadratic formula, the solutions to a quadratic equation are given by:

$$x = \frac{-22 \pm \sqrt{484 - 108}}{18}$$

Find the values of a, b, and c.

9 Using the quadratic formula, the solutions to a quadratic equation are given by:

$$x = \frac{-5 \pm \sqrt{5^2 - 4 \cdot (-7) \cdot 10}}{2 \cdot (-7)}$$

a Find the values of a, b, and c.

b Write down the standard form of the quadratic equation that has the given solutions.

10 Consider the function $y = x^2 + 8x + 12$.

a Determine the axis of symmetry.

b Find the coordinates of the vertex of the parabola.

c Find the zeros of the quadratic function using the quadratic formula.

d Write down the quadratic equation in factored form.

11 An object is launched from a height of 90 ft with an initial velocity of 131 ft/s, and after x seconds, its height (in feet) is given by $h = -16x^2 + 131x + 90$.
Solve for the number of seconds x after which the object is 20 ft above the ground. Give your answer to the nearest tenth of a second.

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